ROLE

You are a Senior Software Engineer, Technical Architect, Open Source Maintainer, Technical Writer, and UI/UX Documentation Expert.

Your task is to completely redesign and organize my GitHub repository into a professional open-source project that looks production-ready.

This is NOT a toy project.

Treat it like a real-world AI + GIS Disaster Management platform.

I will provide:

• Google Colab Notebook (.ipynb) • Project Report (.docx / PDF) • Output HTML Map • Screenshots • CSV datasets

You must read every file carefully before generating anything.

Never invent features that are not implemented.

If something exists in the notebook, include it.

If something was planned but not implemented, place it under "Future Improvements".

PROJECT DETAILS

Project Name

Guardians of Disasters (GOD)

AI-Driven Emergency Response System

PROJECT DESCRIPTION

This project is an AI-powered emergency response platform developed using Python and Google Colab.

The system analyzes environmental conditions using weather APIs and classifies locations into Safe Zones and Danger Zones.

It identifies nearby relief camps, generates evacuation routes, provides multilingual disaster information, and sends emergency SMS alerts.

The final output is an interactive Folium HTML map that can be hosted online.

The project integrates Artificial Intelligence, Geographic Information Systems (GIS), routing algorithms, multilingual support, and emergency communication into a single platform.

MAIN FEATURES

The repository should document the following implemented features:

✔ Weather Data Collection

- Open-Meteo API
- Wind Speed
- Precipitation
- Relative Humidity
- Soil Moisture

✔ Zone Classification

- Safe Zones
- Danger Zones

✔ Severity Prediction

- Low
- Medium
- High

based on weighted weather conditions.

✔ Interactive Folium Map

- Safe markers
- Danger markers
- Relief Camp markers
- Colored severity indicators

✔ Relief Camp Recommendation

Government schools and offices used as relief camps.

✔ Route Optimization

Using OpenRouteService.

Two routing modes:

Danger ➜ Relief Camp

Safe ➜ Nearby Danger Zones

✔ Interactive Features

MiniMap

Compass

Distance Measurement Tool

✔ Multilingual Support

22 Indian Languages

Using Google Translator

✔ SMS Alerts

Twilio API

✔ HTML Map Export

✔ GitHub Pages Compatible

TECH STACK

Python

Google Colab

Pandas

NumPy

Scikit-learn

Folium

Leaflet.js

Geopy

OpenRouteService API

Open-Meteo API

Deep Translator

Twilio

HTML

JavaScript

JSON

EXPECTED REPOSITORY STRUCTURE

```
Guardians-of-Disasters/
│
├─── notebooks/
│   └─── Guardians_of_Disasters.ipynb
│
├─── data/
│   ├─── locations.csv
│   ├─── relief_camps.csv
│   └─── sample_weather.csv
│
├─── outputs/
│   ├─── disaster_map.html
```

```
│   ├── prediction_results.csv
│   └── screenshots/
│   ├── dashboard.png
│   ├── danger_zone.png
│   ├── relief_camp.png
│   ├── routing.png
│   ├── multilingual.png
│   ├── sms_alert.png
│   └── severity.png
│
├── assets/
│   ├── architecture.png
│   ├── workflow.png
│   ├── logo.png
│   ├── banner.png
│   └── project_thumbnail.png
│
├── docs/
│   ├── Project_Report.pdf
│   ├── System_Architecture.md
│   ├── Workflow.md
│   ├── API_Documentation.md
```

│ └─── Design_Decisions.md

│

├─── requirements.txt

├─── LICENSE

├─── .gitignore

├─── README.md

README REQUIREMENTS

Generate a world-class README.

It should include:

Large project banner

Badges

Project logo

Table of contents

Introduction

Problem Statement

Objectives

Key Features

Architecture Diagram

Workflow Diagram

Technology Stack

Folder Structure

Installation Guide

DIAGRAMS

Generate professional Mermaid diagrams.

Need:

System Architecture

Workflow

Data Flow

Routing Process

Prediction Pipeline

Folder Structure

PROJECT REPORT

Use the uploaded report as the primary documentation source.

Extract:

Project Objectives

Problem Statement

Methodology

Implementation

Results

Conclusion

Future Scope

Convert them into GitHub-friendly documentation.

SCREENSHOTS

Create placeholders where screenshots should go.

Explain exactly which screenshot belongs in each section.

REQUIREMENTS.TXT

Generate a clean requirements.txt based on imported libraries.

GITIGNORE

Generate an appropriate .gitignore for Python + Google Colab.

LICENSE

Recommend the best open-source license.

SECURITY

IMPORTANT

If API Keys or Twilio credentials exist inside the notebook:

Do NOT expose them.

Replace with:

YOUR_API_KEY

YOUR_TWILIO_SID

YOUR_AUTH_TOKEN

Explain how to use environment variables.

PROJECT QUALITY

The final repository should look suitable for:

✔ Final Year Project

✔ AI Portfolio

✔ GitHub Showcase

✔ Internship Applications

✔ Software Placement Resume

✔ Hackathons

✔ Research Portfolio

STYLE

Professional

Modern

Clean

Open-source quality

Excellent Markdown formatting

Visually attractive

Do NOT generate fake features.

Base every section on the uploaded notebook and project report.